

DSA 210: Term Project Final Report

Impact of Fuel Prices and Holidays on Flight Ticket Pricing: A Machine Learning Approach

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Introduction to Data Science

1. Introduction & Motivation

The aviation industry is characterized by highly dynamic and complex pricing models. As a student frequently looking for affordable travel options, the seemingly random fluctuations in flight ticket prices served as the primary motivation for this project.

While it is well documented that internal flight characteristics — such as flying Business Class or booking a long-haul flight — drastically increase ticket costs, the impact of external macro-environmental factors is less obvious to the average consumer. Therefore, the core objective of this project was to rigorously investigate two external variables: Global Crude Oil Prices (WTI) and National Holidays.

Initial Hypotheses:

1. **Fuel Cost Dependency:** Since aviation fuel is a massive operational cost for airlines, a spike in global crude oil prices should theoretically trigger an immediate increase in consumer ticket prices.
2. **Holiday Demand Surge:** National public holidays generate massive spikes in travel demand. Basic supply-and-demand economics suggests that flight tickets should be significantly more expensive on these dates.

2. Data Collection, Cleaning, and Enrichment

To conduct this analysis, a singular dataset was insufficient. A rigorous data enrichment process was performed using the pandas and numpy libraries.

2.1 Primary Dataset

The foundational data was sourced from Kaggle's "Flight Price Prediction" dataset, encompassing roughly 300,000 flight records across India. The raw data included columns such as airline, flight, source_city, departure_time, stops, arrival_time, destination_city, class, duration, days_left, and price. This dataset was downloaded as a CSV and served as the baseline for all subsequent merges.

2.2 Data Preprocessing & Cleaning

Several critical transformations were executed in the Jupyter Notebooks:

- **Duration Parsing:** The duration column initially contained string values (e.g., "2h 50m"). A custom Python function was written to extract the hours and minutes via string splitting, converting the entire column into a continuous numeric `duration_minutes` feature.
- **Dropping Irrelevant Features:** The flight code column (e.g., "UK-706") was dropped entirely as it had too high a cardinality and provided no generalized predictive value for the model.

2.3 Data Enrichment via Merging

- **Oil Prices:** Historical WTI Crude Oil daily spot prices were collected by scraping data from Investing.com. Because oil markets are closed on weekends, missing values (NaN) were generated when merging with the flight dataset. The `ffill` (forward fill) interpolation method in pandas was used to carry Friday's oil price over the weekend, ensuring every flight record had an associated fuel cost.
- **Holidays:** A worldwide holiday dataset was acquired and filtered down to relevant national holidays. A binary `is_holiday` feature was engineered by mapping to the flight dates using a left `pd.merge`.

3. Exploratory Data Analysis (EDA) & Hypothesis Testing

Before building predictive models, exploratory data analysis was conducted using matplotlib and seaborn to understand data distributions and test the initial hypotheses.

3.1 Understanding the Baseline: Ticket Price Distribution

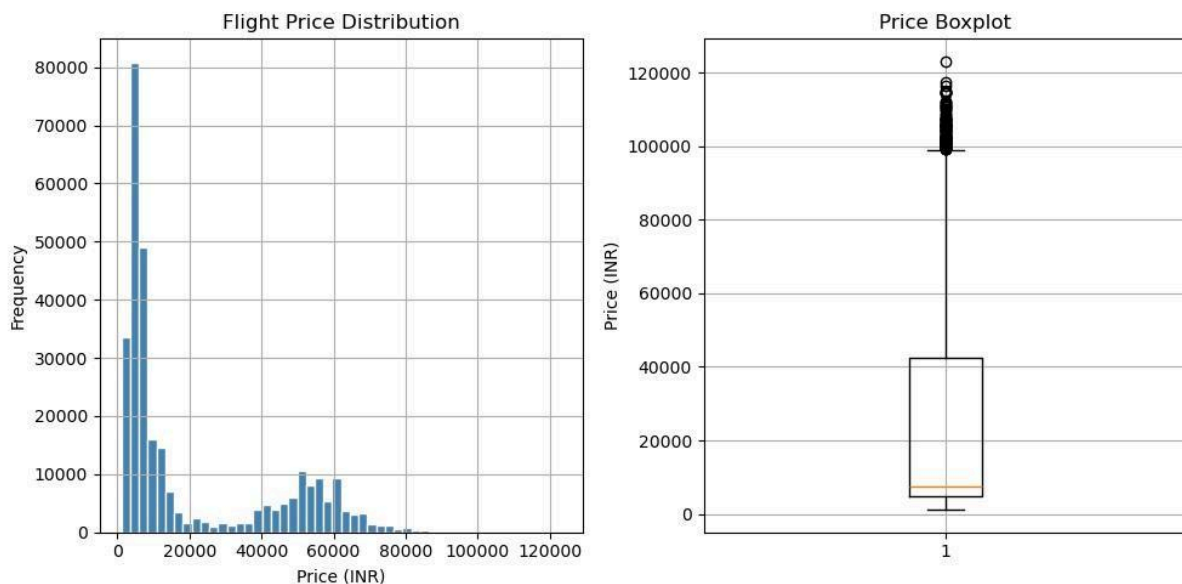


Figure 1 — Flight Price Distribution (histogram) and Price Boxplot

A histogram of the target variable (price) revealed a severe right-skew. The vast majority of ticket prices are clustered in the lower range (Economy class), with a long tail stretching into the upper brackets representing premium seats and last-minute bookings. This skew indicated that a linear model might struggle with extreme outliers.

3.2 Internal Factors: The Class Divide

To baseline the internal factors, average prices were plotted across different airlines segmented by cabin class. The visualization starkly demonstrated that Business class tickets are exponentially more expensive than Economy, regardless of the airline. Vistara and Air India completely dominated the premium market segment.

3.3 Testing Hypothesis 1: The Oil Price Paradox

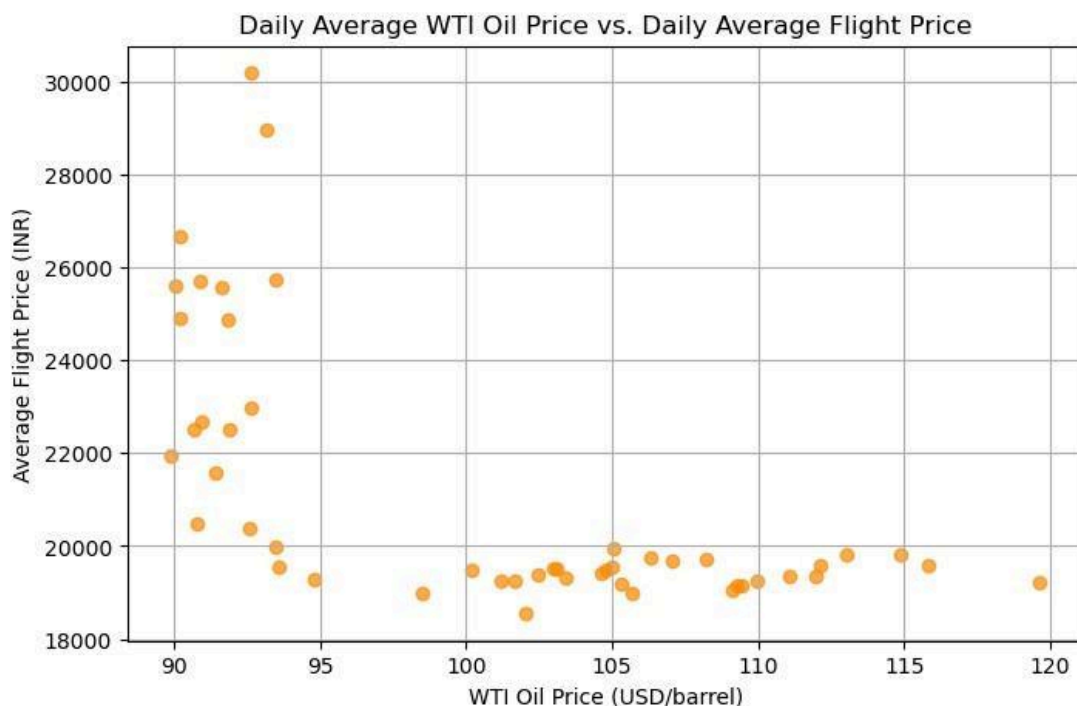


Figure 2 — Daily Average WTI Oil Price vs. Daily Average Flight Price

To test whether higher oil prices correlate with higher ticket prices, the data was grouped by date and daily averages were calculated. `scipy.stats.pearsonr` was used to compute the Pearson correlation coefficient.

Result: The correlation was calculated at $r = -0.6519$. The negative sign meant the hypothesis was entirely wrong — as oil prices went up over the timeframe studied, ticket prices actually trended downwards. Recognising that airlines purchase fuel via "hedging" contracts months in advance, 7-day and 14-day lags were simulated. The correlation weakened ($r = -0.49$ for the 7-day lag), proving that daily spot oil prices do not immediately dictate consumer ticket costs.

3.4 Testing Hypothesis 2: Are Holidays Cheaper?

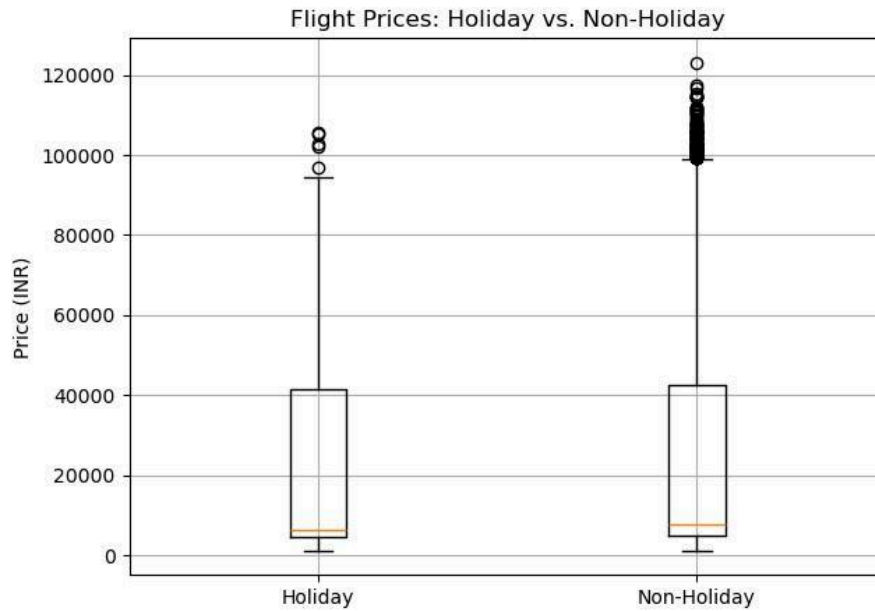


Figure 3 — Flight Prices: Holiday vs. Non-Holiday

To evaluate the holiday demand surge hypothesis, the dataset was split into "Holiday" and "Non-Holiday" groups and a Welch's t-test was run using `scipy.stats.ttest_ind(equal_var=False)`. The p-value was effectively 0.0, indicating statistical significance.

Result: Astonishingly, the mean holiday ticket price was 19,506 INR, while normal days averaged 20,946 INR — flights were statistically cheaper on holidays. The working theory is a demographic shift: highly lucrative corporate and business travel ceases during public holidays, and the influx of cheaper economy leisure travel drags the overall mathematical mean downwards.

4. Machine Learning & Predictive Modeling

In the final phase, the goal was to build predictive models using scikit-learn. The dataset was prepared by applying `pd.get_dummies` to one-hot encode all categorical variables (airline, source_city, destination_city, departure_time, arrival_time, class). Numerical features (duration_minutes, days_left, oil_price) were scaled using `StandardScaler` to prevent larger magnitudes from dominating the loss functions. An 80/20 train-test split was applied.

4.1 The Baseline: Linear Regression

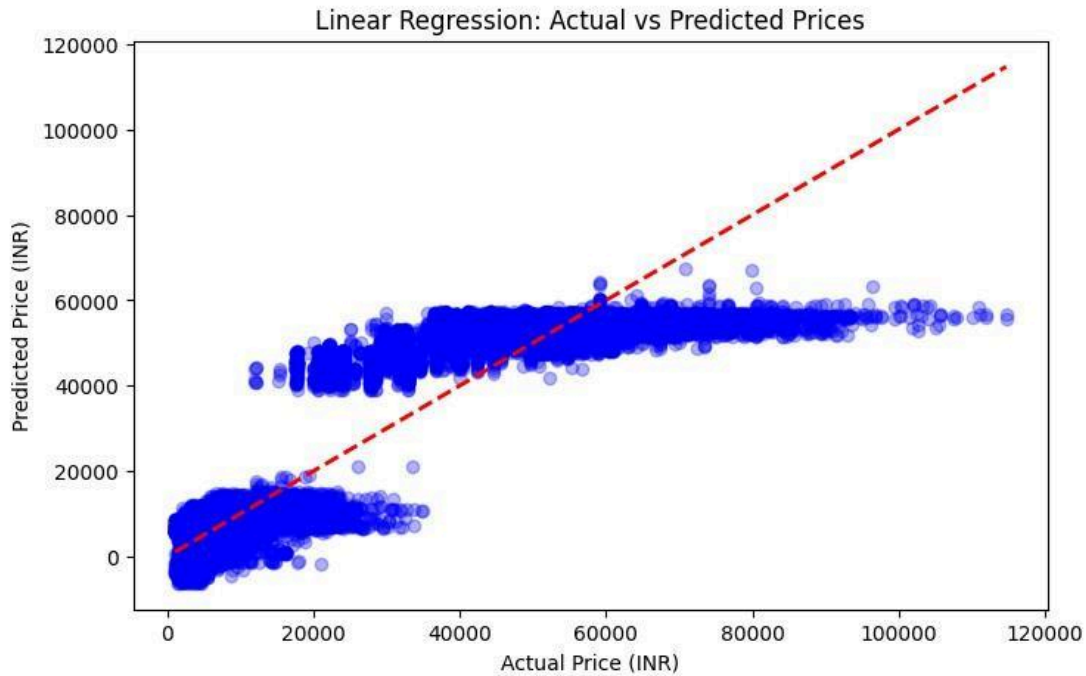


Figure 4 — Linear Regression: Actual vs. Predicted Prices

A standard Linear Regression model was fitted first. It achieved a respectable R^2 score of 0.9049 and a Root Mean Squared Error (RMSE) of 6,995.91 INR. However, as seen in the scatter plot above, the model struggles with higher-priced Business class tickets, forming distinct clusters rather than a cohesive line.

4.2 The Heavyweight: Random Forest Regressor

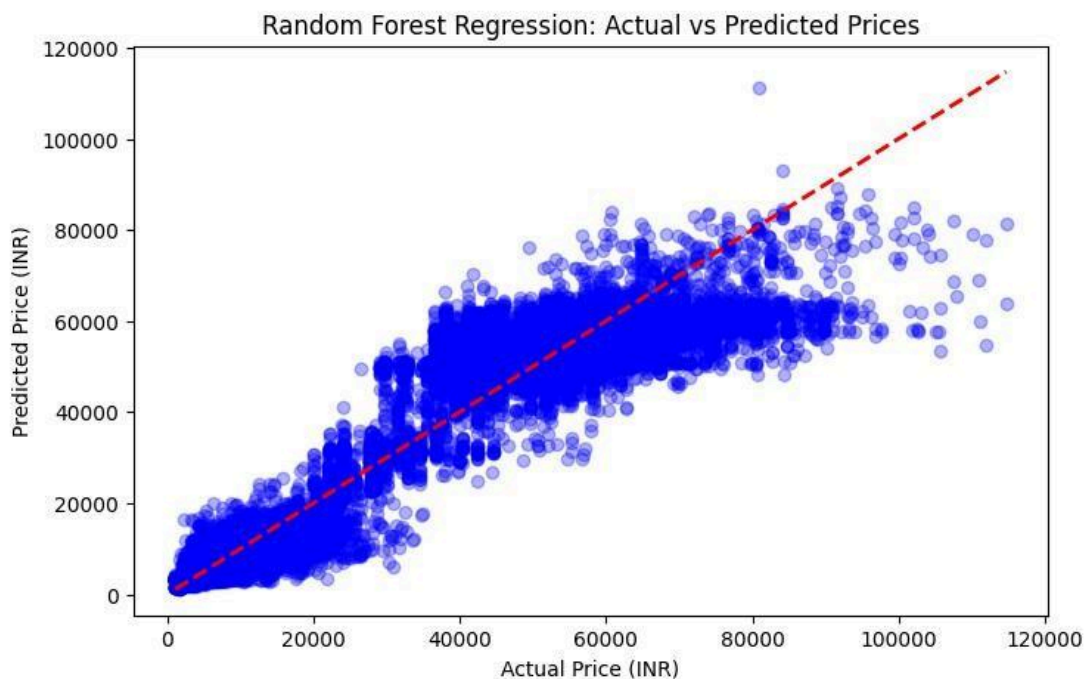


Figure 5 — Random Forest Regression: Actual vs. Predicted Prices

To capture the non-linear tier pricing of airline tickets, an ensemble method was deployed. Given the 300,000-row dataset, running a full Random Forest is computationally expensive. An efficient configuration was therefore adopted, manually tuning the hyperparameters to `n_estimators=50` and `max_depth=15`.

The performance improvement was substantial: the R^2 score jumped to 0.9492 and the RMSE dropped to 5,110.63 INR. The scatter plot shows a much tighter, more accurate clustering along the identity line.

5. Findings: What Really Matters?

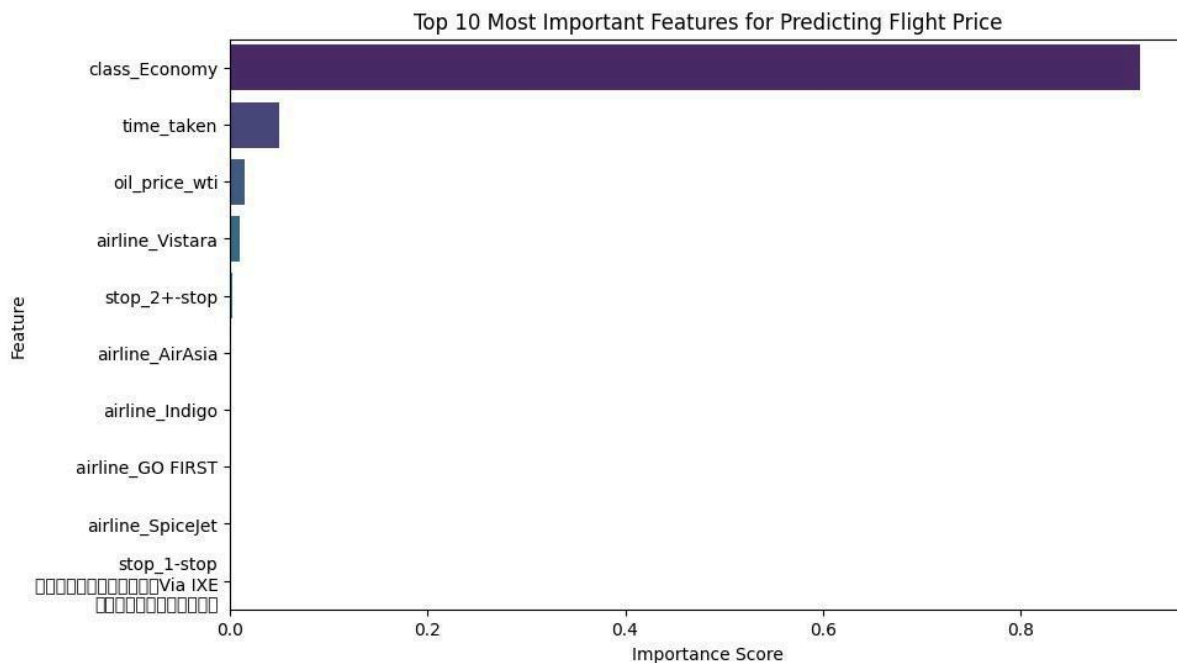


Figure 6 — Top 10 Most Important Features for Predicting Flight Price (Random Forest)

To address the core project question, the `feature_importances_` attribute was extracted from the optimally trained Random Forest model. This metric calculates how much each feature contributes to decreasing the model's impurity (MSE) during training.

The evidence is unambiguous. The model proved that internal flight characteristics absolutely dominate pricing. The binary indicator for `class_Economy` / `class_Business` is by far the most critical predictor, followed by flight duration and the number of days left before departure.

Final Takeaway: While macro-environmental variables like global oil markets and national holidays make for interesting hypotheses, their actual predictive power in the algorithm is virtually zero when compared to basic flight logistics. Airlines price tickets based fundamentally on the seat tier chosen and the distance flown. The data science journey

conclusively disproved both initial hypotheses while simultaneously producing a highly accurate, 94.9% reliable predictive engine.

6. Limitations and Future Work

While the project successfully trained a highly predictive model and rigorously tested the initial hypotheses, several limitations present opportunities for future enhancement:

- **Limited Temporal Scope:** The current dataset captures flight pricing behaviour over a specific, relatively short time window. The inverse correlation between oil prices and flight tickets may therefore be an artefact of this specific period rather than a long-term economic truth. Future work should involve longitudinal datasets spanning several years to better capture cyclical economic trends.
- **Fuel Hedging Nuances:** As discovered during the analysis, airlines use complex hedging strategies to lock in fuel prices months or even years in advance. A future extension could incorporate futures market data or corporate financial reports to map actual fuel expenditure rather than relying on the daily WTI spot price.
- **Time-Series Modelling:** The current approach treats every flight record as an independent observation for regression. However, flight pricing is inherently sequential. Future iterations could explore advanced time-series forecasting architectures such as LSTMs (Long Short-Term Memory networks) or ARIMA models, specifically predicting the price trajectory of a single route as the departure date approaches.
- **Expanded External Variables:** Future analysis could be enriched by incorporating additional macro-environmental factors such as real-time weather conditions at the destination, large-scale sporting or cultural events, or currency exchange rate fluctuations.